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The Influence of Inserts on the Flexural Stiffness of Sandwich Panels

Bachelor's in Mechanical Engineering
Projeto Experimental e Computacional

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Abstract

The aim of this thesis is to determine the influence of through-the-thickness inserts on the flexural stiffness of carbon fibre sandwich panels, a critical structural component in the development of a monocoque chassis for Formula Student. As Formula Student FEUP aims to transition from a tubular frame to a composite monocoque, a thorough understanding of structural connections becomes essential to guarantee mechanical performance and simplify the design process.

To achieve this, a set of specimens were manufactured with embedded inserts and tested, replicating a secondary mounting point in a real monocoque. The lay-up chosen was quasi-isotropic, using prepreg carbon fibre face sheets, an aluminium honeycomb core and aluminium inserts potted with coresplice. The specimens were cured following a three-step cycle using autoclave and oven processes to simulate real-case manufacturing practices.

Flexural stiffness was determined using four-point bending tests in accordance with ASTM D7250/D7250M–06. Results showed that the presence of inserts had no measurable impact on the flexural stiffness of the panel. All specimens presented consistent failure modes, with fibre rupture, core crushing and delamination occurring simultaneously.

To further understand the results, a predictive model was developed using S8R shell elements in Abaqus CAE. The numerical values obtained showed a relative error of 16-17% when compared to the experimental results, which is acceptable given the modelling simplifications. The stiffness overestimation was mainly attributed to the inability of shell elements to account for through-thickness shear deformation.

With this study it is concluded that for low-load applications, inserts do not negatively impact the bending stiffness of CFRP sandwich panels. This allows for local discontinuities to be disregarded in early stage simulations, providing a more efficient design process for FS applications.

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3.1.4 Potting Material

To correctly pot the inserts to the core, FS teams commonly use core splice, an adhesive that expands upon being cured. This material comes in three different forms: resin, film and foam. As discussed in [20], the technique employed needs to account for its application to decide on which form is best.
Considering FS FEUP's need for a production process as simple as possible, the core splice in film or foam form would be the best choice, avoiding unnecessary sizing of injection holes for the insert. Since both forms are viable, to avoid unnecessary purchases, a foam-based core splice was chosen — HexBond ST1150.

3.1.5 Insert

To select the most suitable aluminium grade for the inserts, the ranking weight method was applied once again, taking into account key design requirements. The weighting of each parameter was defined as follows:

- 15% for galvanic potential – Although literature suggests that corrosion is negligible due to the vehicle's short operational lifespan, this study considers potential long-term testing after competition, given the monocoque's role as the team's first. As such, corrosion remains a relevant selection factor;
- 30% for yield stress – Since most inserts are designed to optimize stress distribution, a higher yield strength is essential to minimize weight;
- 25% for manufacturing – The selected aluminium grade must be easy to machine, widely available in European markets, and must not require post-processing heat treatments;
- 30% for cost – Reflecting the team's performance-to-budget objectives.

	Skin material	Corrosion Resistance V	Value 15%	Yield Stress MPa	Value 30%	Manufacturing Grade	Value 25%	Cost €/Kg	Value 30%	Total Value
Aluminium	2024 T3	-0,78	9,4%	290	16%	2	16,7%	3,93 €	11,7%	53,5%
	2014 T6	-0,77	11,3%	324	18%	2	16,7%	3,91 €	14,4%	59,9%
	2017 T4	-0,77	11,3%	244	13%	2	16,7%	2,83 €	30,0%	71,2%
	2219 T87	-0,75	15,0%	331	18%	2	16,7%	4,06 €	13,5%	63,1%
	5052 H32	-0,81	3,7%	152	8%	2	16,7%	3,76 €	15,5%	44,2%
	5083 H111	-0,83	0,0%	131	7%	2	16,7%	3,80 €	15,2%	39,0%
	6061 T6	-0,79	7,5%	240	13%	3	25,0%	3,72 €	15,9%	61,4%
	6063 T6	-0,79	7,5%	205	11%	3	25,0%	3,67 €	16,3%	59,9%
	6060 T6	-0,79	7,5%	150	8%	3	25,0%	3,00 €	25,6%	66,3%
	6082 T6	-0,79	7,5%	240	13%	3	25,0%	3,68 €	16,2%	61,8%
	6066 T6	-0,79	7,5%	300	16%	2	16,7%	7,13 €	5,7%	46,1%
	6013 T6	-0,79	7,5%	324	18%	2	16,7%	3,73 €	15,8%	57,6%
	7075 T6	-0,78	9,4%	460	25%	2	16,7%	7,36 €	5,4%	56,5%
	7050 T7451	-0,78	9,4%	393	21%	1	8,3%	7,39 €	5,4%	44,5%
	7049 T73	-0,78	9,4%	379	21%	1	8,3%	7,35 €	5,4%	43,7%
	7055 T7751	-0,77	11,3%	552	30%	1	8,3%	7,43 €	5,4%	54,9%

Figure 3.3: Weighted Ranking Table for Insert Material Selection

After evaluating several engineering-grade alloys, AW 2017 T4 emerged as the top-ranked material. This choice aligns with industry practices and is supported by its application in other teams, as noted by [20].

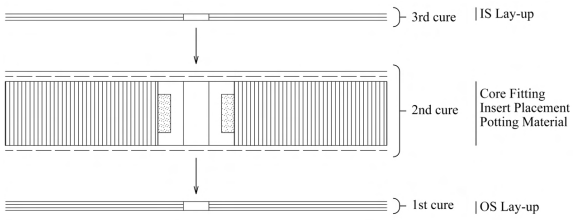


Figure 5.1: Sandwich Panel Manufacturing Steps

border, preventing the centre of the vacuum bag from touching the live edges.

Next began the lay-up of the OS as defined in 4, first all plies were joined together, then holes for the inserts were manually made with punches, after that the panel was layed in the caul plate. It is important to mention that the higher FVF of the prepreg caused fibre distortions in the first $\pm 45°$ ply as seen in 5.4a.

The laminate was then covered by a fluoropolymer release film to prevent undesired material bondings, over it was also placed a sheet of breather cloth with excess material on the desired vacuum valve position to ensure it doesn't seal itself to the surface before pumping all the air out. Afterwards, the actual vacuum bag and valve are added and a five minute drop test was performed to guarantee the bag was sealed. With all steps completed the first cure was performed at $120°C$ at 2.5 bars of pressure for one 1 hour. A higher pressure could not be achieved due to other components being cured simultaneously.

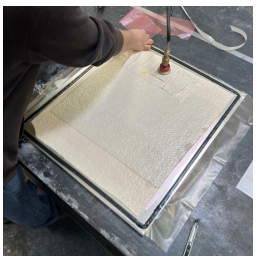


Figure 5.2: Panel Before First Cure

4.2 Insert

4.2.1 Insert Constraints and Load Case

As previously noted, the insert will be sized for a secondary mounting point. To ensure applicability across all non-primary inserts, the heaviest component in this group was selected since the critical loading condition happens when the car is cornering, pulling two times the gravitational acceleration. Therefore the inverter of the car was picked, weighing eight and a half kilograms.

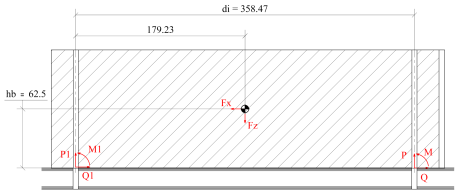


Figure 4.2: Inverter Positioning

The prototype's inverter can be fixed by two inserts as shown in figure 4.2. The load will be applied to inverter's centre of mass, creating a bending moment at the insert. This configuration should guarantee the integrity of the monocoque and pose the biggest geometry changes as possible for this study.

4.2.2 Insert Sizing

To accurately replicate a monocoque structure, a rough sizing of the insert will be performed following [3], the insert design handbook for space standardization. The insert will be evaluated on its capacity to withstand four different loads:

1. Out-of-plane, P;
2. In-plane, Q;
3. Moment, M;
4. Torsion, T - which will not be considered due to the nature of the inverter fixture;

To begin analysis of the insert, the reaction moments and forces at the face sheet hole must be determined. According to [3], since insert variability meets "A" values, a 1.5 safety factor will account for all possible deviations. Both P and Q loads will be divided by both inserts and multiplied by the safety factor, but M must be determined attending to the free-body diagram 4.2:

6 MECHANICAL TESTS

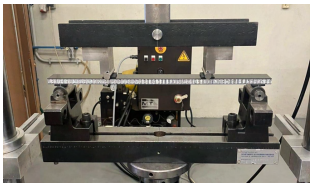
This chapter details the experimental test setup and presents the results obtained on the 31st of May.

6.1 Experimental Set-up

As mentioned in Chapters 2 and 4, to determine the influence of inserts on the flexural stiffness, four-point bending tests with a quarter point load will be performed. The set-up used was: a load span length (L), of 200 mm; support span length (S), of 400 mm; load pin diameter (Ld), of 11.25 mm; support pin diameter (Sd), of 30 mm.

ASTM D7250/D7250M – 06 leaves certain parameters, such as pin diameters, undefined. As the standard references ASTM D7249/D7249M, its recommendations were followed. Accordingly, 25 by 38 mm rubber pads with a thickness of 3 mm and a Shore A hardness of 60 were used, as advised.

The tests were conducted using a universal testing machine equipped with a 10 kN load cell. Only the displacement and reaction force at the load applicator were recorded. For the initial trial, the load applicator displacement rate was set to 1 mm/min, and subsequently increased to 3 mm/min for the remaining specimens.



(a) Testing Set-up Used

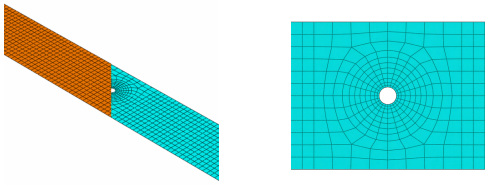


(b) Support and Load Applicator

Figure 6.1: Mechanical Test Set-Up

7.3 Mesh

The sandwich panel was modelled as a continuum shell using S8R elements to accurately capture bending behaviour. All specimens employ a structured mesh with quadratic elements, except for the embedded insert (SI) case. As shown in 7.1b, the presence of the insert hole prevents a structured layout, introducing some element distortion. Nonetheless, the average aspect ratio in the affected region is 1.5, with the most critical element reaching 1.9. These values remain within acceptable limits, as defined in [26], which recommends a maximum ratio of 3 for shell elements.



(a) Simple (Orange) and Insert (Blue) Mesh (b) Mesh Refinement in Insert Specimen

Figure 7.1: Mesh Used for the Simulations

To ensure result reliability, a mesh sensitivity analysis was conducted by comparing the reaction force at the centre node of the right load applicator. As shown in 7.2, variations in element count produced negligible changes in results. For detailed values, please refer to figure E.8.

7.4 Simulation Results and Discussion

To analyse the simulations, the maximum stress criterion was used to determine first ply failure, with data extracted from one hundred integration points in a 3 mm element size simulation. Using both experimental and simulation data, the shear rigidity (U) and bending stiffness (D) were determined according to ASTM D7250/D7250M–06. For the experimental values, an average deflection and maximum load between specimens was considered.

$$U = \frac{G \cdot (d-t)^2 \cdot b}{d-2t} \tag{7.1}$$

$$D = \frac{U \cdot P \cdot (2S^3 - 3SL^2 + L^3)}{24\,(4U\Delta - P(S-L))} \tag{7.2}$$

C.2 Technical Drawing

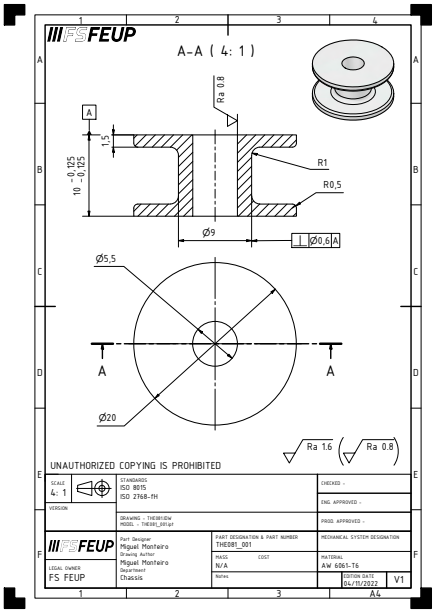


Figure C.3: Insert Technical Drawing - THE081_001

E PREDICTIVE MODELLING AUXILIARY INFORMATION

E.1 Analytical Derivation of Core Properties

The aluminium honeycomb properties were derived based on the following core parameters:

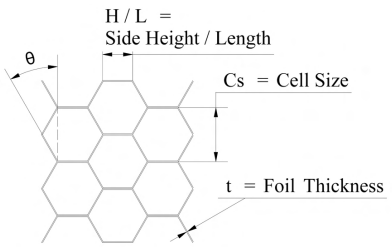


Figure E.1: Honeycomb Parameters

The equivalent in-plane and shear moduli of a honeycomb core can be derived using a micromechanical approach, treating the honeycomb as an array of connected beam elements. This analysis captures the anisotropic behaviour from a cell geometry, and allows for the effective orthotropic elastic constants to be determined.

$$E_1^* = E_s \left(\frac{t}{l_b}\right)^3 \frac{\cos\theta}{(h/l + \sin\theta)\sin^2\theta} \left[\frac{1}{1 + (2.4 + 1.5\nu_s + \cot^2\theta)\left(\frac{t}{l_b}\right)^2}\right] \tag{E.1}$$

$$E_2^* = E_s \left(\frac{t}{l_b}\right)^3 \frac{(h/l + \sin\theta)}{\cos^3\theta} \left[\frac{1}{1 + \left(2.4 + 1.5\nu_s + \tan^2\theta + \frac{2(h_b/l_b)^2}{\cos^2\theta}\right)\left(\frac{t}{l_b}\right)^2}\right] \tag{E.2}$$

E.3 Composite Lay-Up Sections

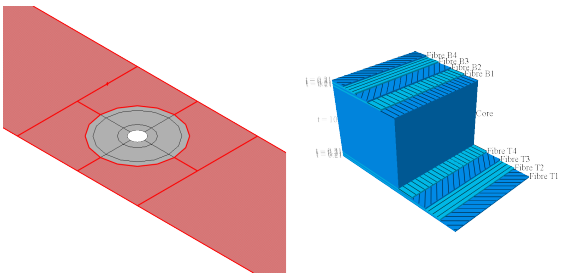


Figure E.4: Section 1 - CFRP and Honeycomb

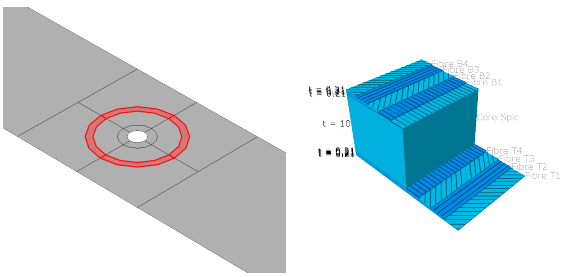


Figure E.5: Section 2 - CFRP and Coresplice